

The Big Picture

Your system converts electricity → RF magnetic field → electricity again, across an air gap. The chain has four physical stages, and efficiency bleeds at every one of them. The whole point of the simulation is to quantify *where* you lose power and *why*, so you can defend every design choice.

The block diagram is:

V_source → TX coil → (air gap, coupling k) → RX coil → L-match → Schottky rectifier → DC load

Each arrow is a stage. Each stage has one job and one efficiency number.

Stage 1 — The Coil Pair and Why 6.78 MHz

You're operating at **6.78 MHz** because it's the AirFuel Resonant (Rezence) standard. This matters for two reasons:

First, at this frequency the coil is electrically small (wavelength ~44 m vs coil diameter ~10 cm), so you're fully in the **near-field regime** — power transfers through a coupled magnetic field, not radiation. This is why Friis transmission equation (far-field) doesn't apply here; the coupling is inductive, not radiative.

Second, 6.78 MHz is an **ISM band** (Industrial, Scientific, Medical), so you're legally allowed to operate without a special licence.

Your coil is modeled with two parameters: **L = 1.5 μ H** (how much magnetic energy it stores) and **R = 0.5 Ω** (copper loss). The ratio of these gives you the **quality factor Q = $\omega_0 L / R \approx 128$** . High Q is good — it means the coil stores energy efficiently. You want $Q > 50$ for reasonable WPT.

The **coupling coefficient k = 0.18** is the key physical parameter for phone charging. It's a number between 0 and 1 that describes how much of the TX coil's magnetic flux actually threads through the RX coil. At $k = 0$, no power transfers. At $k = 1$, the coils are perfectly linked (physically impossible in practice). For a phone flat on a charging pad at ~1 cm separation, $k \approx 0.15$ – 0.25 is realistic.

Mutual inductance M = k·L is what carries the power. The signal at the RX coil is driven by the voltage induced by M: $V_{\text{induced}} = j\omega M \cdot I_{\text{TX}}$.

Stage 2 — Series-Series Resonant Compensation

Without compensation, at 6.78 MHz the coil is mostly a **reactive load** — it looks like $j\omega L = j64 \, \Omega$ to the source. Most of the source power gets reflected, not transmitted. That's terrible for efficiency.

You fix this by adding a **series capacitor** on both TX and RX sides, tuned so that the cap's reactance exactly cancels the coil's reactance at f_0 :

$$C = 1/(\omega_0^2 \cdot L) = 348 \, \text{pF}$$

At resonance, the series LC combination has **zero net reactance** — the coil impedance collapses to just its ohmic resistance $R = 0.5 \, \Omega$. The system "sees" a near-short on each side, which maximises current circulation and therefore maximises the coupled magnetic flux.

This is why the **phase plot** (Figure 1, bottom-left of your dashboard) is important for your presentation — it shows that $\angle Z_{in}$ passes through 0° exactly at 6.78 MHz. Zero phase = purely resistive = resonance achieved. That's your validation that the resonant condition is met.

The **S11 (return loss) plot** validates it from the other side — at f_0 , S11 dips sharply (ideally below $-10 \, \text{dB}$), meaning the source sees a matched load and very little power is reflected back.

Stage 3 — The S-Parameter Model and What It Tells You

This is where your `WPT.m` script lived. The coil pair is modeled as a **two-port network** using Z-parameters (impedance matrix):

- Z_{11}, Z_{22} = self-impedances of each coil loop (including resonant cap)
- $Z_{12} = Z_{21} = j\omega M$ = the mutual coupling term

These are converted to S-parameters because S-parameters directly give you power relationships:

- $|S_{21}|^2$ = fraction of available source power that reaches the RX port. This *is* your coil efficiency η_{coil} . At your operating point, it comes out around 84% at $k = 0.18$.
- $|S_{11}|$ = how much of the source power gets reflected at the TX input. You want this deep in negative dB at f_0 .

The **voltage gain plot** (Figure 1, bottom-right) shows $|V_{\text{out}}/V_{\text{in}}|$ across frequency. It peaks sharply at f_0 , confirming that resonance is not just reducing reflections but actively amplifying the voltage transfer.

Stage 4 — The Thévenin Extraction (The Integration Bridge)

This is the step that was *missing* between your two scripts, and it's the most important part of the integration to explain in your presentation.

At f_0 , looking back into the RX port from the perspective of the matching network, the coil system looks like a **Thévenin equivalent circuit**: a voltage source V_{th} in series with a resistance R_{th} . This is standard Thévenin — any linear two-port network with a source reduces to this.

The formulas come directly from the Z-matrix:

$$V_{th,rx} = Z_{21} \cdot V_{src} / (Z_{11} + Z_0)$$

$$R_{th,rx} = Z_{22} - Z_{12} \cdot Z_{21} / (Z_{11} + Z_0)$$

At resonance, the imaginary parts cancel, and **$R_{th,rx}$ is purely real** — there's a printed check in the console output that verifies Im/Re is essentially zero. This is the whole reason you designed the system at resonance: the matching network problem reduces to a simple **resistor-to-resistor transformation**, not a complex impedance matching problem.

With your parameters, $R_{th,rx}$ comes out around **3.1 Ω** . That's a very low output impedance from the coil. The Schottky rectifier's input impedance is around **500 Ω** . This 160× mismatch is what the L-network has to fix.

Stage 5 — The L-Network Matching

If you just connected the RX coil directly to the rectifier without matching, power would splatter all over the place due to impedance mismatch. The **reflection coefficient** without matching is:

$$\Gamma = (R_{rect} - R_{coil_out}) / (R_{rect} + R_{coil_out}) \approx 0.987$$

That means **97% of arriving power gets reflected back** into the coil, and only ~3% makes it to the rectifier. Catastrophic for a phone charger.

The **L-network** is two reactive elements (one series, one shunt) that transform R_{coil_out} to R_{rect_in} with zero loss at f_0 . The design formula is:

$$Q_{match} = \sqrt{(R_{high}/R_{low} - 1)}$$

With 3.1 Ω and 500 Ω , $Q_{match} \approx 12.7$. That's a high-Q match, which means **narrow bandwidth** — your matching is precise at f_0 but sensitive to frequency drift. The bandwidth of the match is $f_0/Q \approx 0.53$ MHz, which is fine for a narrowband system like this.

The physical elements are a **series inductor (~924 nH)** and **shunt capacitor (~596 pF)**. Both are standard surface-mount values at this frequency.

The **Smith chart** (Figure 5) shows the impedance transformation path. Starting at $R_{\text{coil_out}}$ on the real axis (left side of Smith chart), the series inductor moves the point along a constant-resistance arc clockwise, and the shunt capacitor brings it to $R_{\text{rect_in}}$ on the real axis (right side). Two moves, zero loss. That's the beauty of an L-network.

Stage 6 — The Schottky Rectifier

The Schottky diode does one thing: it only allows current to flow in one direction, converting the AC signal from the coil into pulsating DC. Then the **RC filter** smooths that into steady DC.

You chose a **Schottky diode** (rather than silicon PN junction) because:

- Forward voltage $V_f \approx 0.15 \text{ V}$ (vs $\sim 0.7 \text{ V}$ for silicon) — critical at low RF voltages where 0.7 V is a significant fraction of your signal
- Junction capacitance $C_j \approx 10 \text{ pF}$ — low enough not to short-circuit the RF signal at 6.78 MHz

The half-wave rectifier model gives:

$$V_{\text{dc}} = (V_{\text{peak}}/\pi - V_f) \cdot R_L / (R_L + R_s)$$

The V_{peak}/π term is the average of a half-sine wave. V_f is the diode's entry toll. The $R_L / (R_L + R_s)$ term is a voltage divider between the load and the diode's own series resistance.

The **load sweep plot** (Figure 2) is your rectifier's fingerprint. Efficiency peaks at an **optimal load R_{L_opt}** , then degrades in both directions:

- Too small R_L : most voltage drops across R_s (diode series resistance dominates)
- Too large R_L : power delivered drops as current falls, even though voltage is high

For your report table, η_{rect} peaks around 40–55% depending on the input power level. This is the weakest link in your chain, which is typical — rectifier efficiency is always the hardest to push high at low power levels.

The **power sweep plot** (Figure 3) shows η_{rect} vs. RF input power in dBm. It collapses to zero below a threshold — that's the V_f cutoff: if $V_{\text{peak}} < \pi \cdot V_f \approx 0.47 \text{ V}$, no DC gets through at all. This is the "minimum operating power" floor of your system.

Stage 7 — The Coupling Sweep (Your Phone-Charging Headline)

This is the most practically relevant plot in the whole project (Figure 4), and the one your examiner will ask about.

Every efficiency you've computed so far assumes $k = 0.18$. But a phone doesn't always sit perfectly centred on the pad. As the user shifts the phone, **k drops**. What happens to your DC output?

The sweep shows three things changing with k :

1. **η_{coil} drops** steeply at low k — the coil pair becomes inefficient, most source power is reflected back at TX.
2. **$R_{\text{coil_out}}$ drops** with k (approximately $\propto k^2$) — the RX coil's Thévenin resistance falls, making the mismatch to the rectifier worse unless you re-match.
3. **P_{DC} at the load** drops, and at some minimum k it falls below 5 W — that's your **minimum viable alignment** threshold, marked on the plot.

The key insight to deliver in your presentation: **the coil stage and the matching stage are coupled** — you can't optimise them independently because $R_{\text{coil_out}}$ changes with k . A fixed L-network designed at $k = 0.18$ becomes sub-optimal when the phone shifts. This is why commercial phone chargers use **adaptive matching** or **closed-loop impedance tracking** in the TX controller.

Stage 8 — End-to-End Efficiency (The Bar Chart)

Your final bar chart multiplies the stages:

$$\eta_{\text{total}} = \eta_{\text{coil}} \times \eta_{\text{match}} \times \eta_{\text{rect}}$$

At $k = 0.18$:

- $\eta_{\text{coil}} \approx 84\%$ — near-field resonant coupling does most of the work here
- $\eta_{\text{match}} \approx 100\%$ — ideal L-network, no loss at f_0 (reactive elements only)
- $\eta_{\text{rect}} \approx 45\text{--}50\%$ — this is the bottleneck

$$\eta_{\text{total}} \approx 38\text{--}42\%$$

For 5 W at the phone, you need ~12–13 W from the RF source. With $V_{\text{src}} = 30$ V into $50\ \Omega$, you have ~2.25 W available — which is why at default parameters $P_{\text{DC}} < 5$ W. You'd need to raise V_{src} to ~70–80 V peak (or reduce Z_{source} , which is realistic — a class-E amplifier driving a coil directly has Z_{source} much lower than $50\ \Omega$) to hit the 5 W target.

That gap is honest engineering: it tells you that a real phone charger needs either a higher-power RF driver, a full-wave (bridge) rectifier to double η_{rect} , or a better-matched coil geometry to push k above 0.25.

How to Frame This in Your Presentation

One clean narrative thread: **"We designed a complete RF energy chain and identified efficiency at each stage."**

Open with the block diagram (the image you showed me). Then go through each stage with its efficiency number. Land on the bar chart as your climax — it shows you understand the system end-to-end, not just individual components. Then close with the k-sweep as your real-world validation: "Here's what happens in practice when a user places their phone off-centre, and here's the minimum alignment required."

The three numbers your examiner will ask you to justify are: **$k = 0.18$** (why that value), **$R_{\text{rect_in}} = 500 \, \Omega$** (why that assumption), and **$\eta_{\text{total}} \approx 40\%$** (why so much lower than η_{coil} alone). You now have the physics to defend all three.